VF Asia Footwear Laboratory Testing Manual

Test Method for Fiber Identification	Test method # VF-I02 Page 1 of 1
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PURPOSE AND SCOPE: This test method utilizes chemical analysis to identify the type of polymer or fiber.

REFERENCES: AATCC 20

EQUIPMENT AND MATERIAL:

1. Glass beaker or test tubes
2. Volumetric flasks
3. A balance, capable of weighing to 0.001 grams.
4. 4.4 HCL, reagent grade.
5. Distilled water
6. Thread, fabric, plastic.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Cut 1-2 cm long sample.

REPETITIONS: Test two (2) samples per material**.**

TEST PROCEDURE: (to determine if a thread /yarn is Nylon 6 or Nylon 6.6)

1. Using a glass vessel, prepare a 4.4 N HCL / distilled water solution.

   Note: It is very important that the normality of the solution is exactly 4.4 N. Any variation will produce erroneous results.

2. Cut a 1-2 cm long piece of the thread / yarn , and place it in the bottom of a small glass beaker

3. Add approximately 10 drops of the 4.4 HCl

4. Let the thread / yarn remain submerged in the 4.4 HCl for 10 minutes.

5. After 10 minutes, if the thread / yarn has not dissolved, it is nylon 6.6.

   Note: The 4.4 N HCl may attack / dissolve the bonding agent on the thread /yarn, and subsequently the fibers of the yarn may unravel from each other. This should NOT be interpreted as the thread / yarn dissolving in the HCl

REPORT: Nylon 6 or Nylon 6.6